Geospatial Web Services

Lecture 5

Web Services

- Web services are programming interfaces that can be accessed by other applications over the web

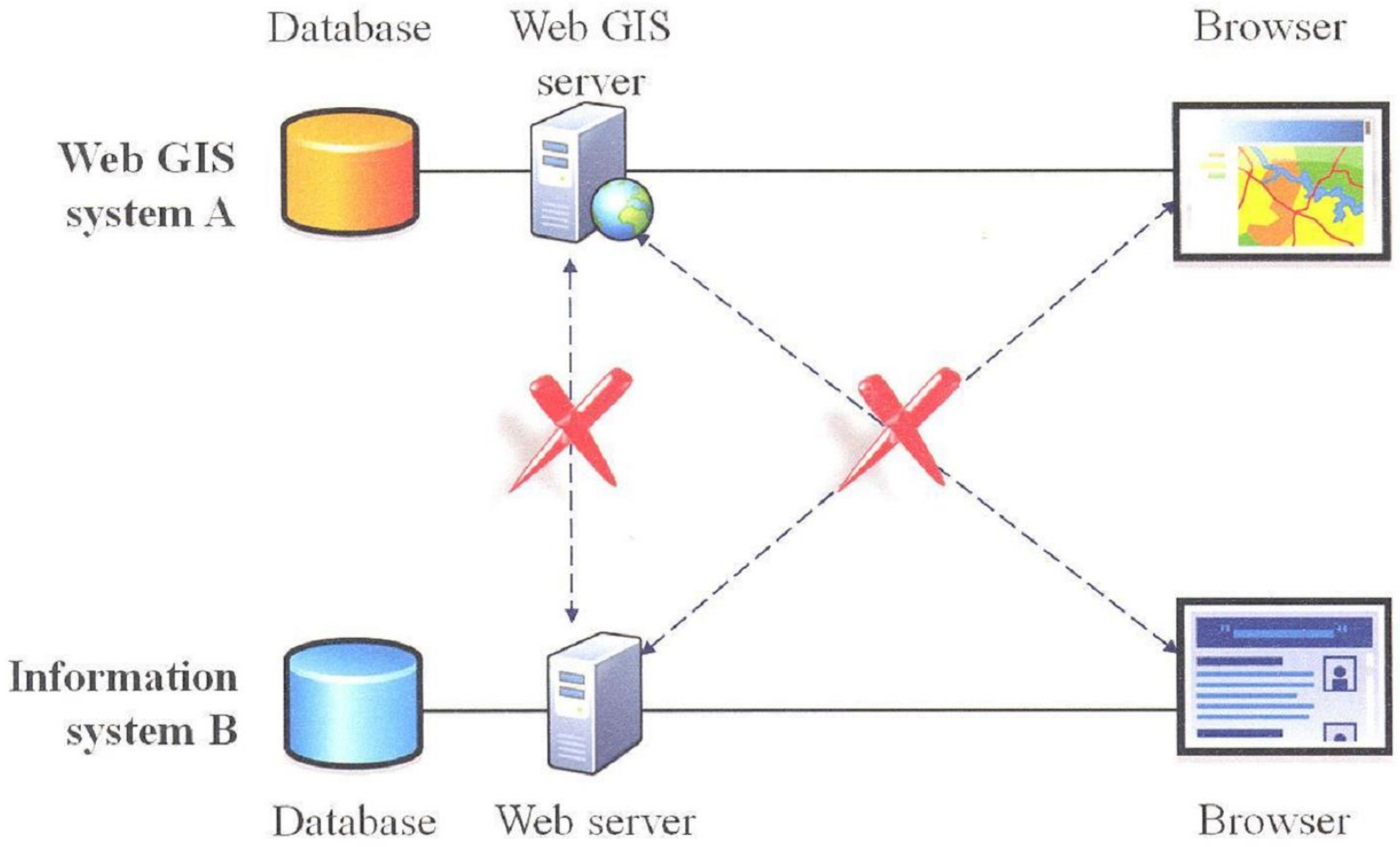
From Website to Web services

- An explosion of web based mapping applications followed the birth of Web GIS in 1993. However, Web GIS was underused, and its potential was not fully utilized, because of

From Website to Web services

- Isolated systems and difficult to reuse and integrate
- Tightly coupled client and server, inflexible to change

From Website to Web services



From Website to Web services

- Web service technology represents an important evolution of distributed computing.
- It replaces a local function with a function that is provided remotely by a web server.

Web pages VS Web services

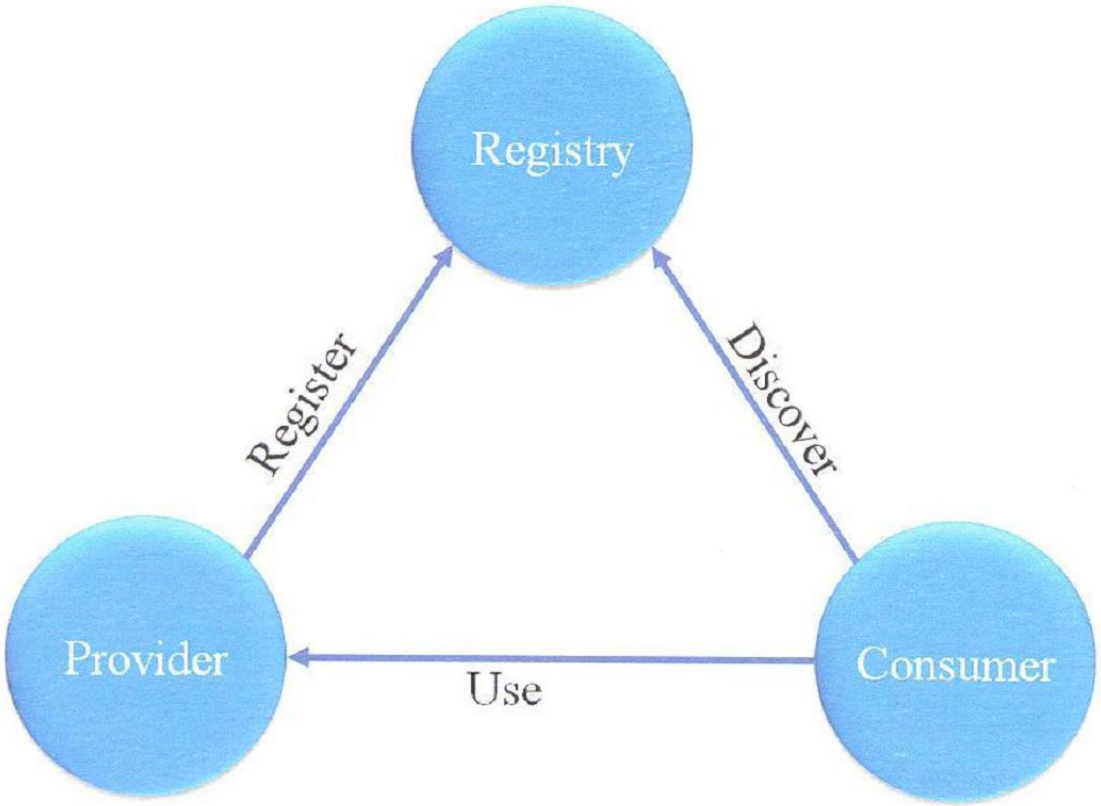
- Web pages are for users to read and understand (HTML)
- Web services are for other computer programs to call

Web services

- There are **three** roles that complete web service architecture.
  - The provider that provides the web service
  - The consumer that consume the web service
  - The registry that acts as a broker between the provider and consumer to facilitate the collaboration

  The provider and consumer don’t necessarily know of each other's existence

Web services



Web services

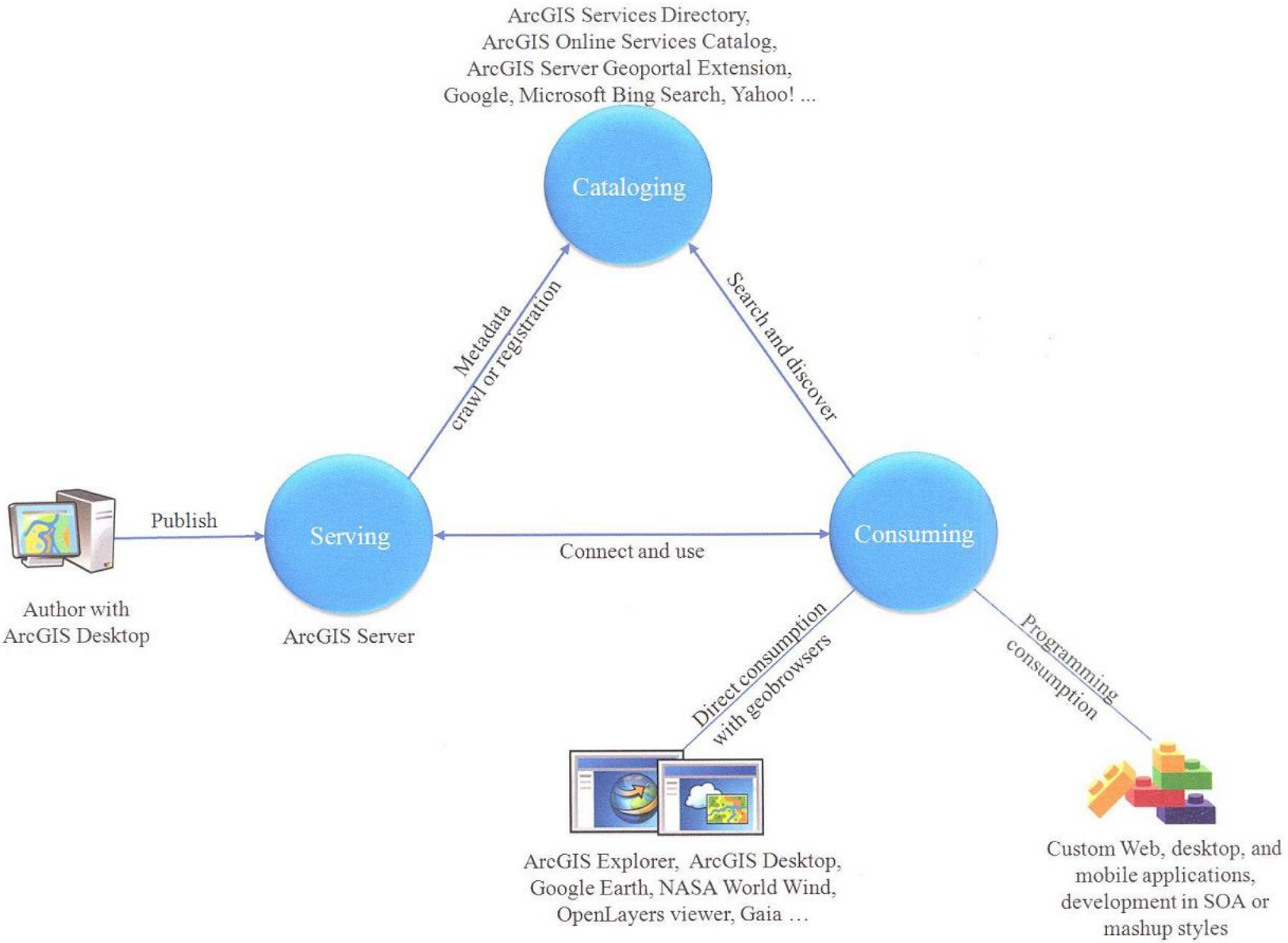
- Web services are programming interfaces and web programs, thus inheriting the characteristics of both

Advantages of Web Services

- Open to other software systems over the web
- Independent of programming language and operating system
- Loosely coupled with remixing in demand
- One-for-all release and update

Geospatial Industry and Support of Web Services

- Corresponding to the provider, consumer, and registry roles in Web services architecture, Web GIS products have evolved into three main categories
  - Server Products (Publisher)
  - Geospatial Web Portals (Registry)
  - Client Products (Consumer)


ArcGIS Services Directory,
ArcGIS Online Services Catalog,
ArcGIS Server Geoportal Extension,
Google, Microsoft Bing Search, Yahoo! ...
Cataloging
Metadata
crawl or registration
Search and discover
Publish
Serving
Connect and use
Consuming
Author with
ArcGIS Desktop
ArcGIS Server
Direct consumption
with geobrowsers
Programming
consumption
ArcGIS Explorer, ArcGIS Desktop,
Google Earth, NASA World Wind,
OpenLayers viewer, Gaia …
Custom Web, desktop, and
mobile applications,
development in SOA or
mashup styles

- Reading Assignment (QUIZ!!)
  - Chapter 3 page 54-61
    - Benefits of Geospatial Web Services
    - Geospatial Web Service Function